Validated Lab Guide
Envelope Modulation of Random Signals

Signal Theory Workshop 02

April 2026

Purpose

This guide is the validated and cleaned deliverable requested by the repository instructions. The refined brief in `docs/lab00_.pdf` is treated as the main source of truth. The result preserves the exact stage names and the same subscript notation used in the original PDF while tightening the structure around parameters, estimators, and required outputs.

Objective

Study how a white Gaussian input process evolves through a conventional AM communication chain:

1. baseband generation by RC low-pass filtering,
2. conventional AM modulation,
3. envelope detection by rectification plus low-pass filtering.

At every stage, estimate the autocorrelation function (ACF) and power spectral density (PSD), then compare the recovered signal with the original message.

Signal Model

Let $x[n]$ be discrete-time white Gaussian noise:

$$x[n] \sim \mathcal{N}(0, 1).$$

The filtered message signal is denoted by $x_1[n]$. The conventional AM signal is

$$x_2[n] = (A_0 + x_1[n]) \cos(2\pi f_0 n T_s),$$

with $T_s = 1/f_s$. The peak and RMS modulation indices are

$$\mu_{peak} = \frac{\max_n |x_1[n]|}{A_0}, \qquad \mu_{rms} = \frac{\sigma_{x_1}}{A_0}.$$

To avoid overmodulation on the observed record, require

$$\mu_{peak} < 1 \qquad \Longleftrightarrow \qquad \min_n (A_0 + x_1[n]) > 0.$$

Under the cycle-averaged interpretation from the lab statement, the AM ACF is

$$R_{x_2x_2}(\tau) \approx \left(\frac{A_0^2}{2} + \frac{R_{x_1x_1}(\tau)}{2} \right) \cos(2\pi f_0 \tau),$$

and the AM PSD contains the carrier and translated sidebands around $\pm f_0$.

Parameter Selection

Use the recommended values from the original PDF, with validated adjustments: the executed notebook uses $A_0 = 6.0$ instead of $A_0 = 4.0$ because the original suggestion can violate $\mu_{peak} < 1$ on a long normalized Gaussian record, and it uses $f_{demod} = 300\,\text{Hz}$ so the discrete-time full-wave detector preserves the recovered envelope while still attenuating the main rectifier ripple around $2f_0 = 600\,\text{Hz}$.

Baseband cutoff	$f_n = 10\,\text{Hz}$
Carrier ratio	$k = 30$
Carrier frequency	$f_0 = kf_n = 300\,\text{Hz}$
Carrier amplitude	$A_0 = 6.0$
Sampling frequency	$f_s = 15{,}000\,\text{Hz}$
Observation time	$T = 50\,\text{s}$
Number of samples	$N = f_sT = 750{,}000$
Demodulator cutoff	$f_{demod} = 300\,\text{Hz}$
PSD method	Welch, Hann window, $N_{seg} = 16384$, 50% overlap

Required Stages

The Colab notebook must preserve these stage headings exactly:

Stage 0: Generation of Input Signal

- Generate $x[n] \sim \mathcal{N}(0, 1)$.
- Estimate and plot $R_{xx}[\ell]$.
- Estimate and plot $S_{xx}(f)$.

Stage 1: RC Low-Pass Filtering

- Filter $x[n]$ with a first-order RC-equivalent low-pass filter.
- Obtain the message signal $x_1[n]$ and normalize it to unit variance.
- Estimate and plot $R_{x_1x_1}[\ell]$ and $S_{x_1x_1}(f)$.
- Compare against the reference RC shapes

$$R_{x_1x_1}(\tau) \propto e^{-2\pi f_n|\tau|}, \qquad S_{x_1x_1}(f) \propto \frac{1}{1 + (f/f_n)^2}.$$

Stage 2: Conventional AM Modulation

- Form

$$x_2[n] = (A_0 + x_1[n]) \cos(2\pi f_0 n T_s).$$

- Verify $\mu_{peak} < 1$ and report μ_{rms}.
- Plot the AM waveform with its envelope.
- Estimate and plot $R_{x_2x_2}[\ell]$ and $S_{x_2x_2}(f)$.

Stage 3: Envelope Detection

- Rectify: $x_{rect}[n] = |x_2[n]|$.
- Low-pass filter the rectified waveform to obtain $x_{env}[n]$.
- Remove the DC component and normalize the result to obtain $x_{demod}[n]$.
- Estimate and plot $R_{demod}[\ell]$ and $S_{demod}(f)$.
- Compute

$$\rho = \mathrm{corr}(x_1, x_{demod}), \qquad \mathrm{NMSE} = \frac{\mathbb{E}\big[(x_1 - x_{demod})^2\big]}{\sigma_{x_1}^2}, \qquad \mathrm{SNR} = -10\log_{10}(\mathrm{NMSE})\ \mathrm{dB}.$$

Comparison and Analysis

- Compare $x_1[n]$ and $x_{demod}[n]$ in time and frequency.
- Compare $R_{x_1x_1}[\ell]$ with $R_{demod}[\ell]$.
- Compare $S_{x_1x_1}(f)$ with $S_{demod}(f)$.
- Discuss sidebands, envelope positivity, and demodulation error.

Required Deliverables

The submission must contain:

1. the executed notebook,
2. this PDF guide,
3. the following figures inside the notebook:
   - Stage 0 ACF and PSD,
   - Stage 1 ACF and PSD with theoretical overlays,
   - Stage 2 waveform, ACF, and PSD,
   - Stage 3 detection and recovery plots,
   - modulation-index analysis with four subplots,
   - error analysis with four subplots,
   - comprehensive comparison with a 2×2 layout.
4. the reported metrics:
   - μ_{peak} and μ_{rms},
   - ρ,
   - NMSE,
   - SNR in dB,
   - power efficiency

$$\eta = \frac{\mu_{rms}^2}{1 + \mu_{rms}^2} \times 100\%.$$

Validation Notes

This guide intentionally keeps the original notation and stage names from the lab PDF while making the contract explicit:

- the AM chain is conventional AM, not DSB-SC,
- the detector is full-wave rectification plus low-pass filtering,
- the comparison signal is $x_1[n]$,
- the recovered output is $x_{demod}[n]$ after DC removal and normalization.